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Ethical Usage of AI

AI became an essential part of 74% of professionals' weekly work. Most of them would agree it's been helping them boost productivity and save great amounts of time in their professional lives/careers. However, frequent AI usage also comes with different risks such as plagiarism, bias or different privacy violations. All of these things are just minor risks nowadays. As AI evolved, so did the risks of using it. While discussing the ethics of AI, it's important to focus on fairness, transparency, accuracy, privacy, etc.

Among the major problems mentioned in the article is that AI uses datasets that can include biased, inaccurate, or even harmful information. As a result, the algorithm makes biased decisions and gives inaccurate results. Transparency is another problem mentioned in the article. Users cannot understand the sources of information used by the system and the reasons why AI made certain decisions. Thus, users cannot trust the AI algorithm or use its suggestions.

Human supervision is very important for the process of using AI algorithms. Despite the fact that artificial intelligence can perform many tasks faster than humans, it does not guarantee the high quality of the work. People should not become fully dependent on such technology and use it blindly. This is an indication that organizations have to develop some regulations and guidelines related to the use of AI algorithms.

Many different concerns are raised in this article, all of which are closely entwined with software engineering. One of the most notable of these concerns is privacy and security. At many different software engineering companies, the use of AI can pose privacy

and security concerns on multiple different fronts. On one level, the posting of code and information to AI chatbots might expose sensitive information. At companies where safe AI usage isn't explicitly discussed, programmers usually end up paving their own path, leading to poor and uninformed decision making on what is considered sharable and not.

Additionally, the code the AI writes itself can lead to privacy and security breaches. This is especially true in agentic programming, where AI strategizes and plans out entire software systems, and then implements them all at once. This kind of AI programming is especially prone to improper code, as at least when developers prompt AI to implement specific functions, they need to manually copy paste or accept the output. However, when agentially coding, massive amounts of lines are produced at once. Without an exhaustive system to check code that is produced, this naturally leads to significant chunks of unreviewed code entering a code base.

For example, infamously, an unsecured, misconfigured legacy cloud storage bucket exposed 72,000 files from accounts from the Tea App. This app had its code completely vibe coded, and obviously the firebase configurations were part of the code that never went reviewed. This is an extreme example, as this app was built by someone with obviously zero programming experience. However, it illustrates how a lack of proper review of code can lead to security and privacy breaches.

Solutions to issues of ethical AI usage are numerous. One of the most important things I think discussed in the article regarding ethical AI usage is "Role specific-training." If someone is using AI, they should have knowledge of when to trust, verify, and override AI outputs. If users are not confident in their understanding of what an AI is generating, then that person shouldn't be in a position to be using AI for that task.

I think that this is extremely important, because without this baseline, it becomes extremely hard to attribute blame for AI failures. It is both unsustainable for your company

and employees. If employees lie about understanding and verifying code that their AIs generate, then when an AI generates incorrect code they will be forced to be held accountable. Additionally, if a company doesn't enforce stringent policy on the usage of AI and the qualifications necessary, then they will face complete liability if something goes terribly wrong as a result of employee negligence, for it isn't necessarily the employees fault if the company just never explicitly states rules regarding AI usage.

I think it is actually an extremely interesting question. Is it reasonable to offload complete responsibility of AIs outputs to the AI/AI company itself? This is already done in some realms, and it is considered reasonable. For example, Waymos are automatic cars that run without human intervention. Even though Waymos aren't perfect, and occasionally crash, they are still far less likely to crash than a human is. For that reason people are willing to put absolute trust in Waymo, without feeling the need to sit in the driver's seat.

I think it is reasonable that at a certain point we will reach a similar level with AI code. However, the next big question, which is covered in the article, is risk classification. For developing a frontend website, offloading complete responsibility onto an agentic AI is a reasonable decision because the worst thing that can happen is a poorly planned website. However, a lot more is hinging on for example the coding of a nuclear reactor, or the system architecture for a moon landing apparatus. The risk inherent to failure of these systems could be catastrophic, to the point where even a 1% chance of faulty code is not acceptable.

In conclusion, the debate around the application of AI to software engineering is an over growing and changing field. But, as it currently stands, when using an AI to do something for you, you should ask the questions, "Do I have the ability to trust, verify, and override what my AI is telling me?" If you can't answer yes, you probably should do more research to better understand roughly what you are doing. Additionally, your willingness to trust AI should depend on the risk classification of what you are doing. If you are coding your

authentication system and password storage, blindly trusting AI is probably going to work, but if it doesn't, the damage associated with failure is massive. Are you willing to trust AI for something so crucial? By asking yourself your competence on the subject matter, and assessing the risk associated with an incorrectly implemented system, you can make more ethical decisions on your AI usage.

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